

TicketChain — Week 1 Progress Report

CS414 Fundamentals of Blockchain · Localized P2P Event Ticketing Network

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1. Summary

During Week 1 we established the full project skeleton for TicketChain, a decentralized peer-to-peer event ticketing network that represents tickets as ERC-721 NFTs and prevents scalping through an on-chain price ceiling. The repository now contains four working areas — **frontend/**, **contracts/**, **network/**, and **docs/** — each with an initial, runnable implementation.

2. Repository Setup

- Repository structured into **frontend/** (React UI), **contracts/** (Hardhat + Solidity), **network/** (PyIPv8 P2P layer), and **docs/** (architecture & reports).
- Hardhat project initialized inside **contracts/** with OpenZeppelin (ERC-721) dependencies and a Chai/Mocha automated testing framework configured for next week's contract test work.
- Python virtual environment isolated inside **network/** with a pinned **requirements.txt** (pyipv8 2.13, cryptography 43.0.3).
- System architecture diagram added to **docs/architecture.md** and linked from the root README.

3. Smart Contracts

- Drafted **TicketNFT.sol**: an ERC-721 contract where each ticket is minted by the event organizer with a hard-coded face value (0.05 ETH).
- Stubbed the custom **transferTicket** function with the basic face-value price guard in place; full anti-scalping escrow and settlement logic is scheduled for Week 2.
- Wrote an initial Chai/Mocha suite covering minting, access control, face-value transfers (sunny day) and above-face-value rejection (rainy day).

4. P2P Network Layer

- Implemented **main.py** boilerplate that boots an IPv8 instance running our custom **TicketChainCommunity** overlay.
- Configured a local development overlay using UDP-broadcast bootstrapping, keeping discovery isolated from public IPv8 trackers.
- Verified with **test_two_nodes.py**: two instances on ports 8090/8091 discover each other in under 2 seconds.

5. UI Creation

- React + Vite + TypeScript app scaffolded in **frontend/**, deployed at **ticket-chain-ten.vercel.app**.
- Wallet connectivity wired through wagmi/viem; initial components built: Navbar, TicketCard, and TicketGrid for browsing available tickets.

6. Current Challenges

- Dependency compatibility: pyipv8 2.13 breaks against the newest *cryptography* release (removed legacy EC curves); resolved by pinning cryptography 43.0.3.
- Designing the escrow flow for **transferTicket** so payment settlement and NFT transfer happen atomically — main contract task for Week 2.
- Bridging on-chain events into the PyIPV8 overlay (listener service design still open).
- WSL2 networking quirks when testing UDP broadcast discovery across machines (works locally).

7. Plan for Week 2

- Implement the full anti-scalping transfer logic (escrow + settlement) in TicketNFT.sol.
- Expand the contract test suite (concurrency, edge cases, gas measurements).
- Connect the frontend to the locally deployed contract via wagmi.
- Prototype ticket-availability gossip messages inside TicketChainCommunity.